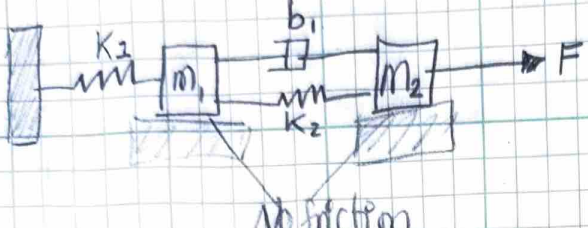
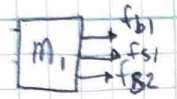
2.1(c) Write differential equations for systems. State whether they will decay, given nonzero initial conditions.

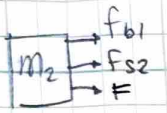




$\sum F = ma$

$ma = f_{s1} + f_{b1} + f_{s2}$

$m_1\ddot{x}_1 = -K_1x_1 + b_1(\dot{x}_1 - \dot{x}_2) + K_2(x_1 - x_2)$



$\sum F = ma$

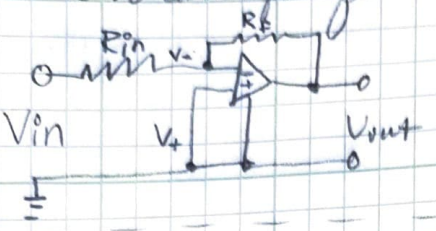
$ma = f_{b1} + f_{s2} + F$

$m_1\ddot{x}_1 = b_1(\dot{x}_1 - \dot{x}_2) + K_2(x_1 - x_2) + F$

The system will decay to rest.

(2.13) $V_{out} = \frac{10^7}{S+1}[V_+ - V_-]$; $i_+ = i_- = 0$

find transfer function



$$\frac{V_{in} - V_n}{R_{in}} = \frac{V_n - V_{out}}{R_f}$$

$$V_{in}(R_f) - V_n(R_f) = V_{out}(R_{in}) - V_n(R_{in})$$

$$V_{out} = \frac{10^7}{S+1}[0 - V_-] = \frac{-V_n(10^7)}{S+1}$$

$$V_{out}(S+1) = -V_n(10^7)$$

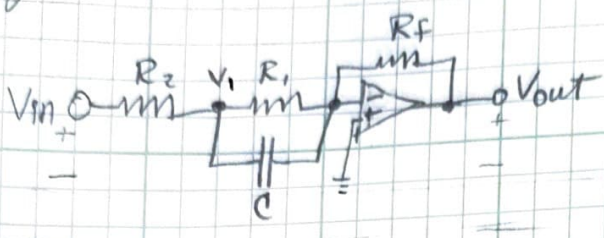
$$V_n = \frac{V_{out}*(S+1)}{10^7}$$

$$V_{in}(R_f) - \frac{(V_{out})(S+1)}{10^7}(R_f) = V_{out}(R_{in}) - \frac{(V_{out})(S+1)}{10^7}(R_{in})$$

$$V_{in}(R_f) = V_{out}(R_{in}) - \frac{(V_{out})(S+1)}{10^7}(R_{in}) + \frac{(V_{out})(S+1)}{10^7}$$

$$\boxed{\frac{V_{out}}{V_{in}} = \frac{-R_f}{R_{in} - \frac{(S+1)(R_{in})}{10^7} + \frac{S+1}{10^7}}}$$

(2.17b) Write dynamic equations and find transfer function



$$i_{R2} = i_{R1} + i_C = i_{Rf}$$

$$\frac{V_{in}-V_1}{R_2} = \frac{V_1-0}{R_1} + C\frac{dV_1}{dt} = \frac{0-V_{out}}{R_f}$$

$$\frac{V_{in}-V_1}{R_2} = \frac{-V_{out}}{R_f} \Rightarrow (V_{in}-V_1)R_f = (-V_{out})R_2$$

$$V_{in}R_f - V_1R_f = -V_{out}R_2$$

$$V_{in}R_f + V_{out}R_2 = V_1R_f$$

$$V_1 = \frac{V_{in}R_f + V_{out}R_2}{R_f} = V_{in} + \frac{V_{out}R_2}{R_f}$$

$$\frac{V_{in} + \frac{V_{out}R_2}{R_f}}{R_1} + C\frac{d}{dt}\left[\frac{V_{in}+V_{out}R_2}{R_f}\right] = \frac{-V_{out}}{R_f}$$

$$\frac{V_{in}}{R_1} + \frac{V_{out}R_2}{R_1R_f} + C\frac{d}{dt}\left[\frac{V_{in}+V_{out}R_2}{R_f}\right] = \frac{-V_{out}}{R_f}$$

$$\frac{V_{in}}{R_1} + \frac{V_{out}R_2}{R_1R_f} + C\frac{d}{dt}\left[\frac{V_{in}}{R_f}\right] + C\frac{d}{dt}\left[\frac{V_{out}R_2}{R_f}\right] = \frac{-V_{out}}{R_f}$$

$$\frac{V_{in}}{R_1} + \frac{V_{out}R_2}{R_1R_f} + \frac{C}{R_f}\frac{d}{dt}[V_{in}] + \frac{CR_2}{R_f}\frac{d}{dt}[V_{out}] = \frac{-V_{out}}{R_f}$$

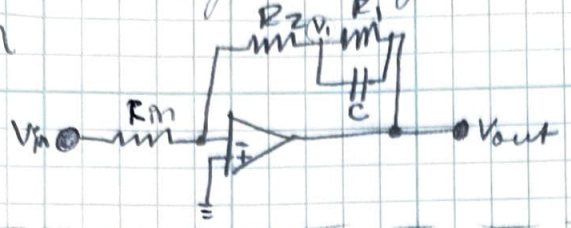
$$\frac{V_{in}(s)}{R_1} + \frac{V_{out}(s)R_2}{R_1R_f} + \frac{C}{R_f}(s)V_{in}(s) + \frac{CR_2}{R_f}(s)V_{out}(s) = \frac{-(s)V_{out}(s)}{R_f}$$

$$\frac{V_{in}(s)}{R_1} + \frac{C}{R_f}(s)V_{in}(s) = \frac{-V_{out}(s)R_2}{R_1R_f} - \frac{CR_2}{R_f}(s)V_{out}(s) - \frac{(s)V_{out}(s)}{R_f}$$

$$V_{in}\left[\frac{s}{R_1} + \frac{C}{R_f}(s)\right] = V_{out}\left[\frac{-sR_2}{R_1R_f} - \frac{CR_2s}{R_f} - \frac{s}{R_f}\right]$$

$$\frac{V_{out}(s)}{V_{in}(s)} = \frac{\frac{s}{R_1} + \frac{C}{R_f}(s)}{\frac{-sR_2}{R_1R_f} - \frac{CR_2s}{R_f} - \frac{s}{R_f}}$$

2.17(c) Write dynamic equations and find transfer function



$$\frac{V_{in}-0}{R_{in}} = \frac{0-V_1}{R_2} = \frac{V_1-V_{out}}{R_1} + C\frac{d}{dt}\left[V_1-V_{out}\right]$$

$$\frac{V_{in}}{R_{in}} = \frac{-V_1}{R_2} \rightarrow -\frac{V_{in}R_2}{R_{in}} = V_1$$

$$\frac{V_{in}}{R_{in}} = -\frac{V_{in}R_2}{R_{in}} + C\frac{d}{dt}\left[-\frac{V_{in}R_2}{R_{in}} - V_{out}\right]$$

$$\frac{V_{in}}{R_{in}} = -\frac{V_{in}R_2}{R_{in}} + \frac{-CR_2}{R_{in}}\frac{d}{dt}\left[V_{in}\right] - \frac{d}{dt}\left[V_{out}\right]$$

$$\frac{V_{in}(s)}{R_{in}} = -\frac{V_{in}(s)R_2}{R_{in}} - \frac{CR_2}{R_{in}}(s)V_{in}(s) - (s)V_{out}(s)$$

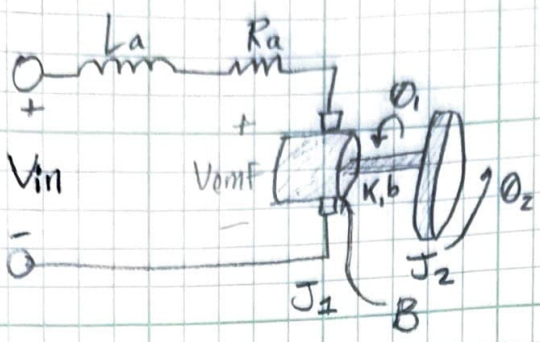
$$\frac{V_{in}(s)}{R_{in}} + \frac{CR_2}{R_{in}}(s)V_{in}(s) + \frac{V_{in}(s)R_2}{R_{in}} = -(s)V_{out}(s)$$

$$V_{in}(s)\left[\frac{1}{R_{in}} + \frac{CR_2(s)}{R_{in}} + \frac{R_2}{R_{in}}\right] = V_{out}(s)\left[-(s)\right]$$

$$V_{in}(s)\left[\frac{CR_2(s)+R_2+1}{R_{in}}\right] = V_{out}(s)\left[-s\right]$$

$$\boxed{\frac{V_{out}(s)}{V_{in}(s)} = -\frac{CR_2(s)+R_2+1}{R_{in}(s)}}$$

(2.22)



An electric motor driving load that had one dominant vibration mode.

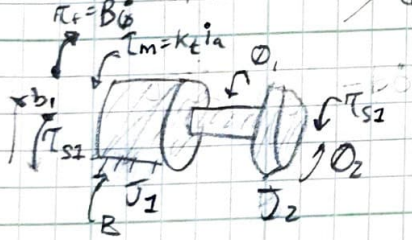
motor electrical constant = K_e
torque constant = K_t
armature inductance = L_a
resistance = R_a
motor inertia = J_1
viscous friction = B
load inertia = J_2

Two inertias are connected by shaft w/ spring constant K and an equivalent viscous damping b.

Write the equations of motion.

$V_{emf} = K_e \dot{\theta}$

$$V_{in} - R_a i_a - L_a \frac{di_a}{dt} - V_{emf} = 0 \xrightarrow{\mathcal{L}} V_{in}(s) - R_a I_a - K_e s\Theta(s) = 0$$



$\Sigma T = I\alpha$

$J\ddot{\theta} = T_m - T_f - T_{s1}$

$J_1\ddot{\theta}_1 = K_t i_a - B_1\dot{\theta}_1 - K(\theta_1 - \theta_2) - b(\dot{\theta}_1 - \dot{\theta}_2)$

$J_1 s^2\Theta_1(s) = K_t I(s) - B_1 s\Theta(s) - K(\Theta_1(s) - \Theta_2(s)) - bs(\Theta_1(s) - \Theta_2(s))$

$J_2\ddot{\theta}_2 = T_{s1}$
$K(\theta_1 - \theta_2) - b(\dot{\theta}_1 - \dot{\theta}_2)$
$= K(\Theta_1(s) - \Theta_2(s)) - bs(\Theta_1(s) - \Theta_2(s))$

$J_1 s^2\Theta_1(s) = K_t I(s) - B_1 s\Theta(s) - K(\Theta_1(s) - \Theta_2(s)) - bs(\Theta_1(s) - \Theta_2(s))$

$J_2 s^2\Theta_2(s) = K(\Theta_1(s) - \Theta_2(s)) - bs(\Theta_1(s) - \Theta_2(s))$

$$V_{in} - R_a i_a - K_e\dot{\theta}_1 = 0$$